



Java Operator Precedence

[< Previous](#)[Next >](#)

Java Operator Precedence

When a calculation contains more than one operator, Java follows **order of operations** rules to decide which part to calculate first.

For example, multiplication happens before addition:

Example

[Get your own Java Server](#)

```
int result1 = 2 + 3 * 4;    // 2 + 12 = 14
int result2 = (2 + 3) * 4;  // 5 * 4 = 20

System.out.println(result1);
System.out.println(result2);
```

[Try it Yourself »](#)

Why Does This Happen?

In `2 + 3 * 4`, the multiplication is done first, so the answer is `14`.



Tip: Always use parentheses () if you want to make sure the calculation is done in the order you expect. It also makes your code easier to read.

Order of Operations

Here are some common operators, from highest to lowest priority:

- () - Parentheses
- * , / , % - Multiplication, Division, Modulus
- + , - - Addition, Subtraction
- > , < , >= , <= - Comparison
- == , != - Equality
- && - Logical AND
- || - Logical OR
- = - Assignment

Another Example

Subtraction and addition are done from left to right, unless you add parentheses:

Example

```
int result1 = 10 - 2 + 5;    // (10 - 2) + 5 = 13
int result2 = 10 - (2 + 5); // 10 - 7 = 3

System.out.println(result1);
System.out.println(result2);
```

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

Remember: Parentheses always come first. Use them to control the order of your calculations.

Exercise [?]

Drag and drop the correct operator to add the numbers.

```
int sum = 5      3;
```

+ - * /

[Submit Answer »](#)

Video: Java Operators

[Tutorials](#) ▼[References](#) ▼[Exercises](#) ▼[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)[< Previous](#)[Sign in to track progress](#)[Next >](#)

ADVERTISEMENT

[Tutorials](#) ▼[References](#) ▼[Exercises](#) ▼[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

COLOR PICKER

[REMOVE ADS](#)

ADVERTISEMENT





GET CERTIFIED

FOR TEACHERS

FOR BUSINESS

CONTACT US

Top Tutorials

- HTML Tutorial
- CSS Tutorial
- JavaScript Tutorial
- How To Tutorial
- SQL Tutorial
- Python Tutorial
- W3.CSS Tutorial
- Bootstrap Tutorial
- PHP Tutorial
- Java Tutorial
- C++ Tutorial
- jQuery Tutorial

Top References

- HTML Reference
- CSS Reference
- JavaScript Reference
- SQL Reference
- Python Reference
- W3.CSS Reference
- Bootstrap Reference
- PHP Reference
- HTML Colors
- Java Reference
- AngularJS Reference
- jQuery Reference

Top Examples

- HTML Examples
- CSS Examples
- JavaScript Examples
- How To Examples
- SQL Examples
- Python Examples
- W3.CSS Examples
- Bootstrap Examples
- PHP Examples
- Java Examples

Get Certified

- HTML Certificate
- CSS Certificate
- JavaScript Certificate
- Front End Certificate
- SQL Certificate
- Python Certificate
- PHP Certificate
- jQuery Certificate
- Java Certificate
- C++ Certificate

[Tutorials](#) ▼[References](#) ▼[Exercises](#) ▼[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)[FORUM](#) [ABOUT](#) [ACADEMY](#)

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning.

Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our [terms of use](#), [cookie and privacy policy](#).

[Copyright 1999-2025](#) by Refsnes Data. All Rights Reserved. [W3Schools](#) is Powered by [W3.CSS](#).